



SANDRO RODRÍGUEZ MUÑOZ

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Graduate student in Physics with an MSc in Mathematics, passionate about research, learning, and teaching. My academic interests lie in the foundations of mathematics and science, with a particular focus on category theory and its applications to physics and other disciplines. I am deeply motivated by interdisciplinary approaches that connect abstract mathematical structures with physical theories, aiming to contribute to both their theoretical understanding and pedagogical development.

EDUCATION

Complutense University of Madrid (UCM)

09/2019 - 02/2024

Bachelor's Degree in Physics

Erasmus at the LMU in Munich (last year of the degree)

Data Analysis with Python (40-h Course) (2023-2024)

University of Granada (UGR)

09/2024-09/2025

Master's in Mathematics

PROGRAMMING AND SOFTWARE SKILLS

Programming Languages

Python, JavaScript, MATLAB

Python Packages

Pandas, Matplotlib, NumPy, Keras

Software & Tools

HTML, LaTeX, Excel, COMSOL

EXPERIENCE

Research Associate (UGR)

02/2025 - 07/2025

Practicum at the UGR

- Conducted an in-depth literature review on “Categorical Quantum Mechanics” for the Department of Algebra at the University of Granada.
- Wrote a survey document explaining the most relevant concepts of the field, its history, its current trends and its most relevant articles, with a special focus on the book “Categories for Quantum Theory: An Introduction”, by Chris Heunen and Jamie Vicary.
- Made a guide to navigate the extensive literature on the topic, and compiled the most important articles and books in one place.
- Gave a presentation to interested members of the department, explaining the main ideas and developments of the subject.

Master's Thesis (UGR)

02/2025 - 09/2025

Categorification of QFT

- Wrote an extensive thesis on the formulation of Quantum Field Theory (QFT) using category theory, analyzing the recent developments on the topic made by authors such as Stefano Gogioso, Fabrizio Genovese, etc.
- Introduced the main physical concepts needed for QFT, “translating” the physical arguments into a more suitable format for mathematicians.
- Gave an overview of Categorical Quantum Mechanics and how its concepts are used and generalized for the categorical formulation of QFT.

Bachelor's Thesis (UCM)

02/2023 - 06/2023

Machine Learning and Autonomous Vehicles

- Wrote a brief (20-page limit) thesis on how machine learning algorithms can be used to make self-driving vehicles, and made 2 implementations using different methods and tools.
- Explained the main concepts of machine learning relevant for the thesis, their mathematical formalization, and how they are implemented in the case of autonomous vehicles.
- Implemented genetic algorithms using JavaScript to create a virtual car that after various generations was able to drive successfully in different traffic situations.
- Constructed a simple neural network in Python and trained it to be able to drive a virtual car on a more realistic 3D simulated road.

Volunteering Work

09/2023 - Present

Acción en Red Andalucía (NGO)

- Conducted tutoring sessions and educational workshops for young students from vulnerable backgrounds.
- Organized charity events and public awareness campaigns focused on social issues.
- Designed and implemented educational programs for children and high school students.
- Provided support for victims of violence through awareness initiatives, debates, and community outreach.
- Participated in talks and presentations representing the organization.

Tutoring

09/2023 - Present

Self-Employed

- Provided private tutoring in Mathematics, Physics, Chemistry, English, and German for pre-university students.
- Delivered university-level tutoring in Statistics and Physics.
- Offered both online and in-person classes adapted to diverse learning styles and needs.
- Developed personalized teaching materials and learning resources.
- Prepared students for official exams and university entrance examinations.

REFERENCES

José Gómez Torrecillas: Professor at the Department of Algebra, University of Granada (UGR).

Email: gomezj@ugr.es

Website: [José Gómez Torrecillas](#)

María Varela del Arco: Professor at the Department of Materials Physics, Complutense University of Madrid (UCM); Director of the ICTS–National Center for Electron Microscopy.

Email: mvarela@fis.ucm.es

Website: [María Varela del Arco](#)

LANGUAGES

Spanish: Native.

English: C2 Certificate of Proficiency in English Cambridge.

German: B2 Goethe-Zertifikat.